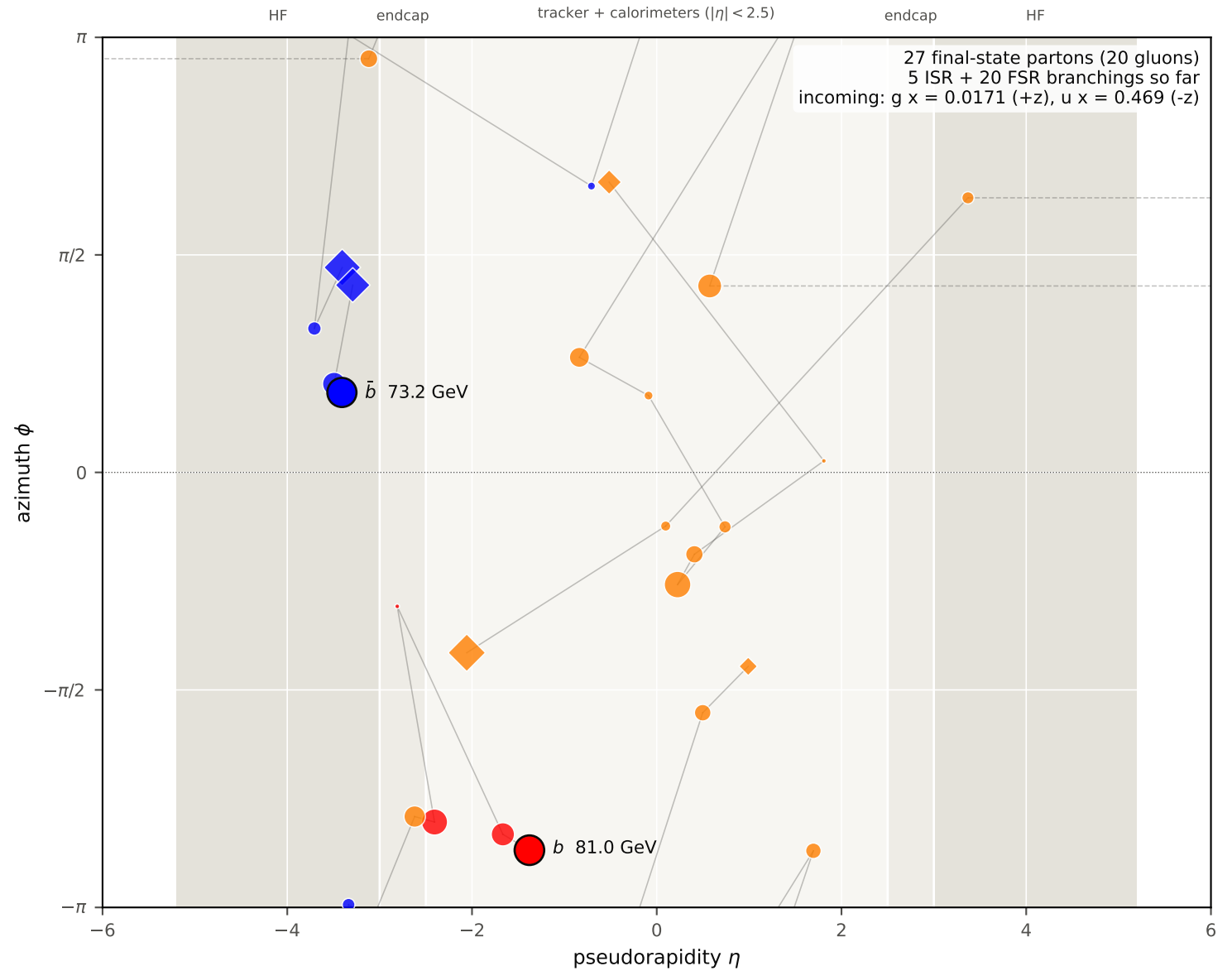
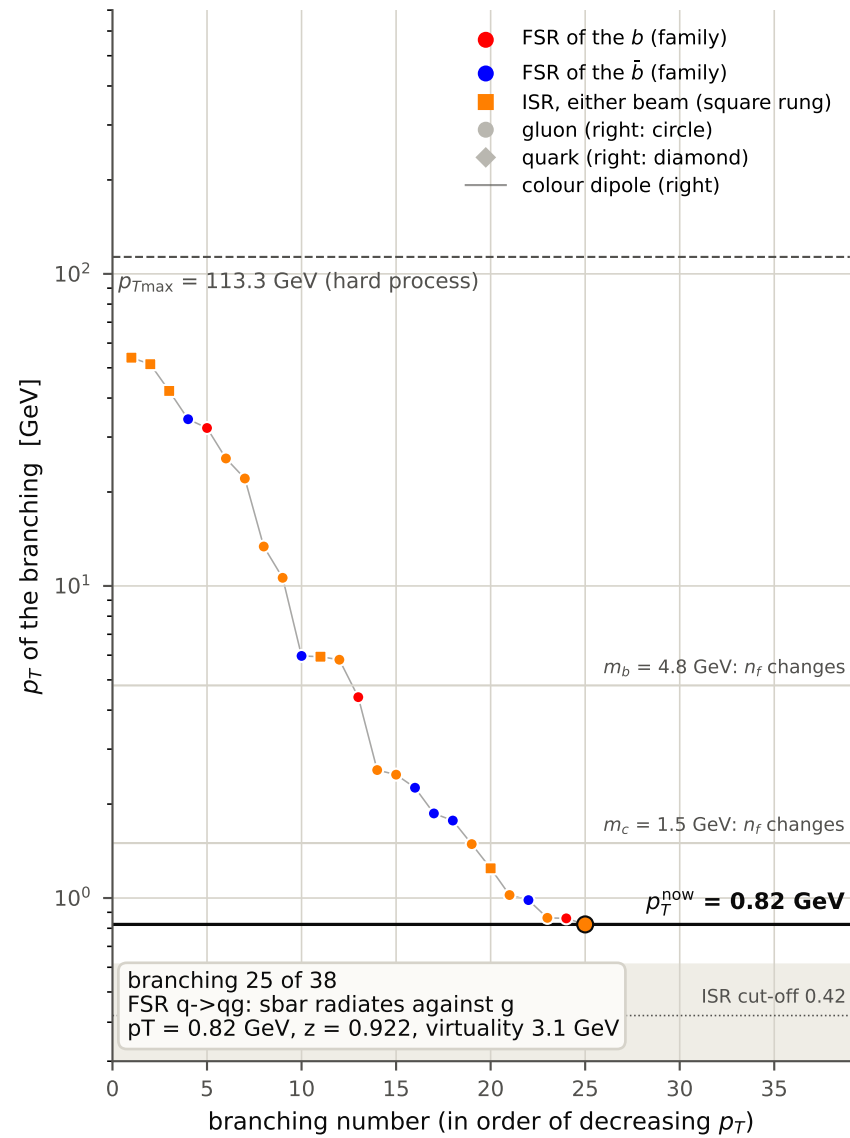


The shower unwinds in p_T : branching by branching from 113 GeV down to the cut-off

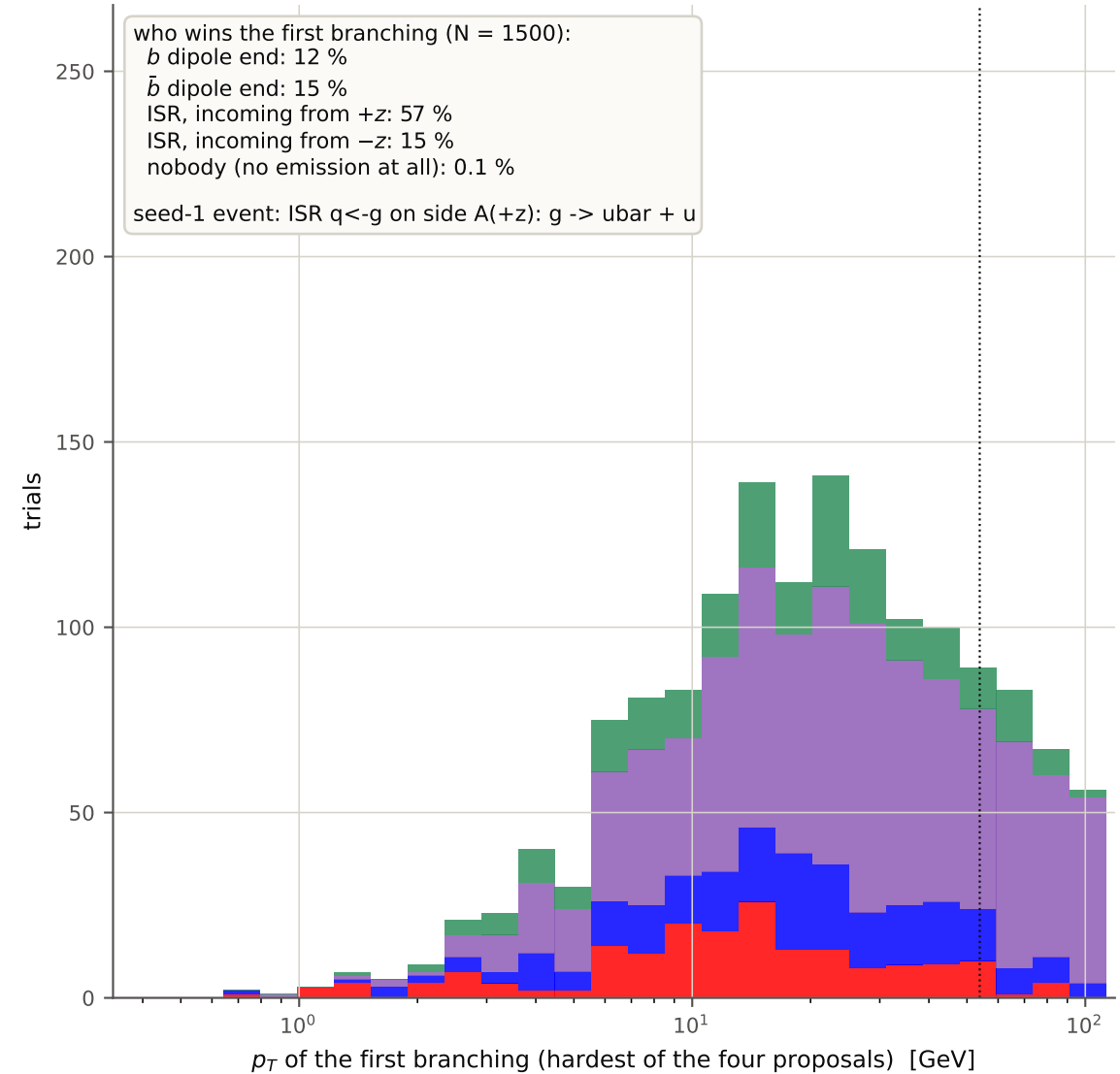
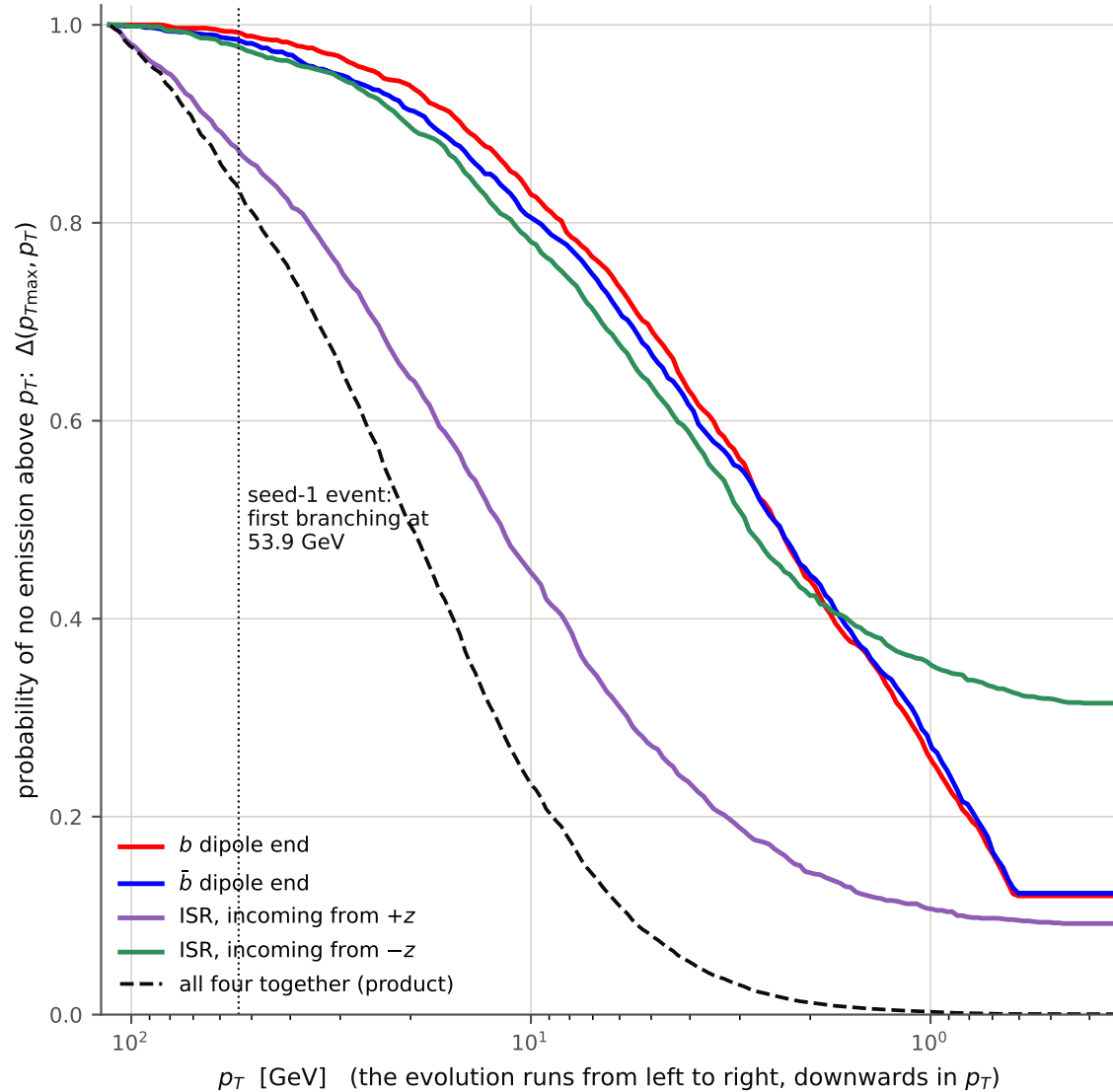


Lower in p_T the emissions get softer and more frequent: α_s grows and $1/p_T^2$ takes over. A gluon may also split into $q\bar{q}$, here even into $b\bar{b}$.

Who radiates first? The Sudakov veto algorithm as an ensemble of proposals

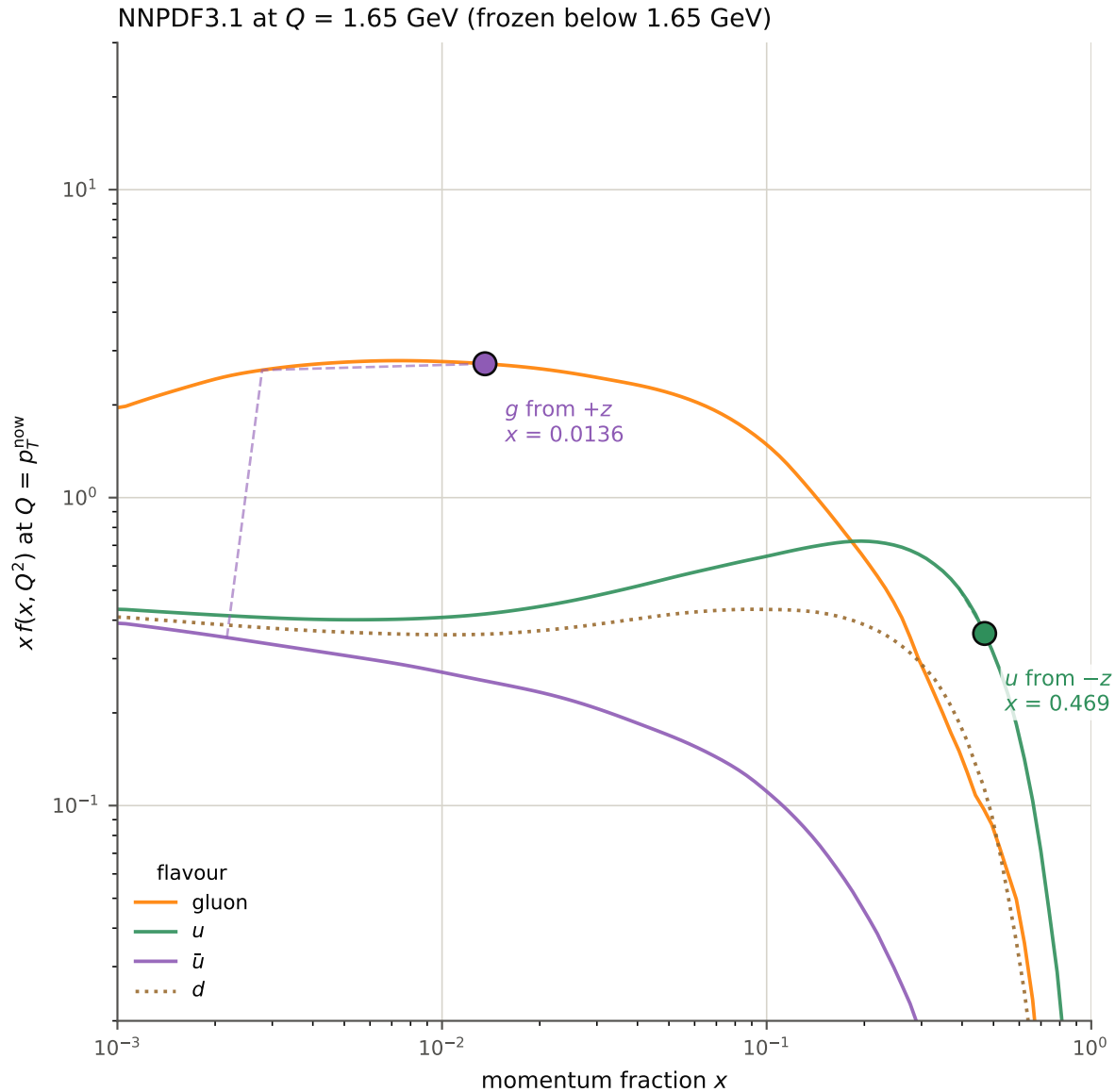
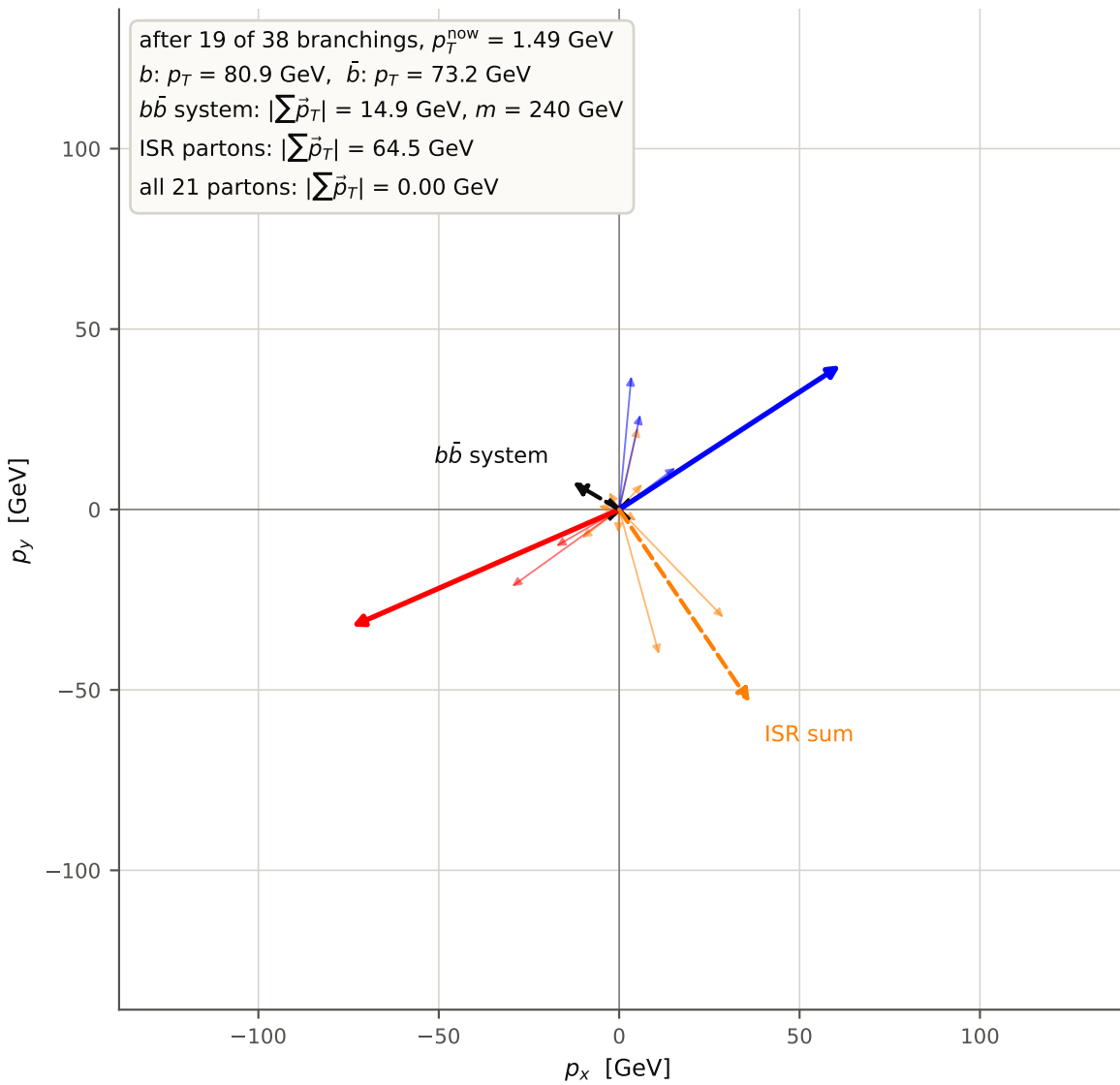
scene 2/4

$N = 1500$ proposals per end, from $p_{T\max} = 113.3$ GeV



The hardest proposal wins: here ISR, incoming from +z does in 57 % of the trials, because the PDF ratio $x'f(x')/xf(x)$ is large at small x .

Momentum bookkeeping: the $b\bar{b}$ system recoils against the ISR, the beams walk up in x

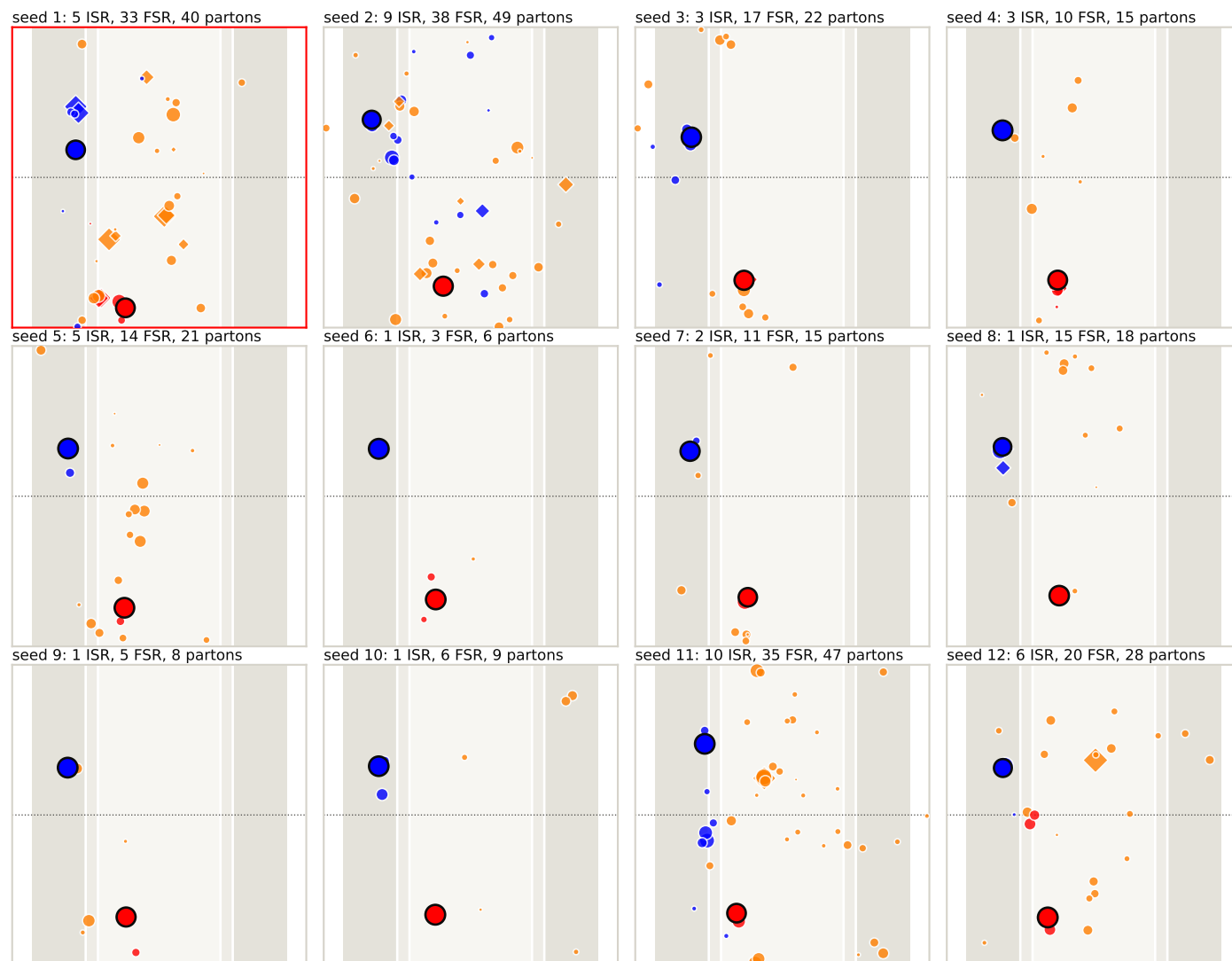


Backward evolution: each ISR branching replaces the incoming parton by its mother at x/z , a step up in x , while the PDFs themselves change with the falling scale.

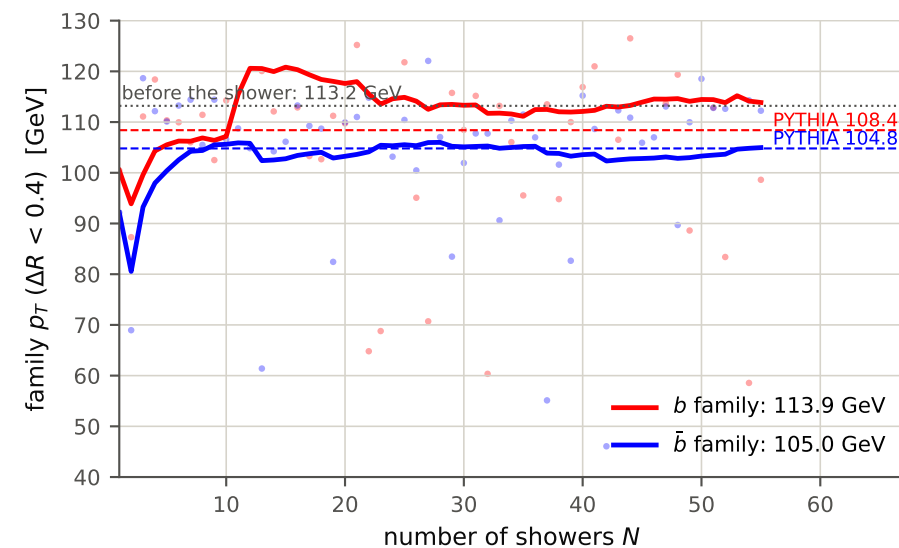
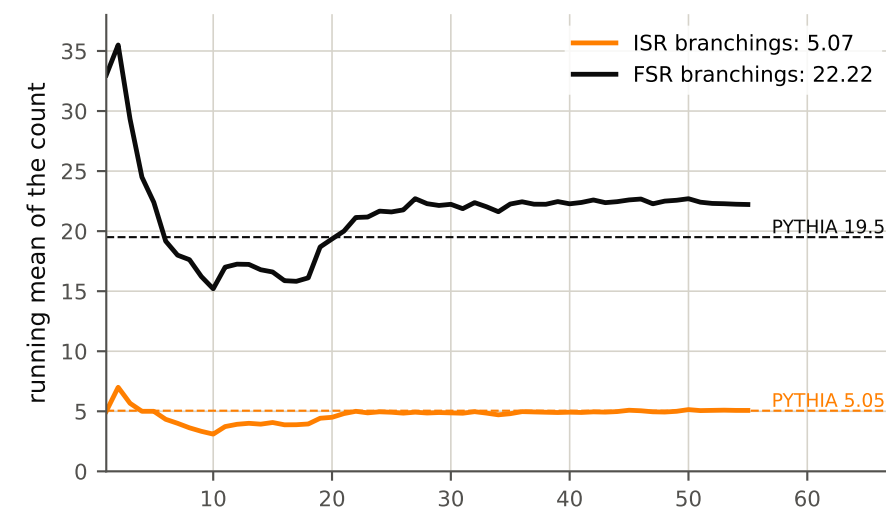
The same hard event showered again and again: no two alike, the averages agree with PYTHIA

scene 4/4

(η, ϕ) of the final state, seeds 1 to 12 (the book's event is seed 1)



$N = 55$ showers of the same hard event



After 64 showers: 5.2 ISR, 22.2 FSR branchings, family p_T 113 / 106 GeV; PYTHIA 8 on the same event: 5.05 / 19.5 and 108 / 105 GeV.